

MAT101R01 – ENGINEERING MATHEMATICS – I

(Common for all branches of Engineering) Semester I

Unit IV: Improper and Multiple Integrals

Improper integrals – Beta and Gamma integrals, their properties and relation – Duplication formula – Evaluation of definite integrals in terms of Beta and Gamma integrals – Double integrals – Change of order of integration – Triple integrals – Area, volume, mass and moments of inertia.

Lecture Notes by **Dr. S. Balaji****Contents**

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1 Improper integrals

Definition 1.1 (Improper integral)

A definite integral $\int_a^b f(x) dx$ is said to be an **improper integral** of the

- (i) **first kind**, if either the limit a or b or both are infinite;
- (ii) **second kind**, if $f(x)$ is unbounded at one or more points of the interval $[a, b]$;
- (iii) **third kind**, if it is of both the first and second kinds.

Examples. (1) $\int_0^\infty e^{-x} dx$ (or $\int_1^\infty \frac{1}{x^2} dx$) is an improper integral of the first kind.

(2) $\int_0^1 \frac{dx}{1-x^2}$ is an improper integral of the second kind (the integrand is unbounded at $x = 1$).

(3) $\int_0^\infty e^{-x} x^{n-1} dx$ (with $0 < n < 1$) is an improper integral of the third kind.

2 Gamma integral

Definition 2.1 (Gamma integral)

An improper integral of the form $\int_0^\infty e^{-x} x^{n-1} dx$ ($n > 0$) is called the **Gamma integral** and is denoted by $\Gamma(n)$, i.e.

$$\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx, \quad n > 0.$$

Result 2.1 (Properties of the Gamma integral)

- (1) $\Gamma(1) = 1$
- (2) $\Gamma(n+1) = n\Gamma(n)$
- (3) $\Gamma(n+1) = n!$ if n is a positive integer

Proof of (1). By definition $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$

$$\Rightarrow \Gamma(1) = \int_0^\infty e^{-x} x^0 dx = \int_0^\infty e^{-x} dx = \left(\frac{e^{-x}}{-1} \right)_0^\infty = -(0 - 1) \Rightarrow \boxed{\Gamma(1) = 1}$$

Proof of (2). By definition $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$

$$\Rightarrow \Gamma(n+1) = \int_0^\infty e^{-x} x^{(n+1)-1} dx = \int_0^\infty e^{-x} x^n dx.$$

Integrate by parts, $\int u dv = uv - \int v du$, with $u = x^n$, $dv = e^{-x} dx$, so $du = nx^{n-1} dx$, $v = -e^{-x}$:

$$\begin{aligned} \Gamma(n+1) &= -\left(x^n e^{-x}\right)_0^\infty - \int_0^\infty (-e^{-x}) nx^{n-1} dx \\ &= -(0 - 0) + n \int_0^\infty e^{-x} x^{n-1} dx = n\Gamma(n) \Rightarrow \boxed{\Gamma(n+1) = n\Gamma(n)} \end{aligned}$$

Here $\lim_{x \rightarrow \infty} x^n e^{-x} = \lim_{x \rightarrow \infty} \frac{x^n}{e^x}$ is of the form $\frac{\infty}{\infty}$; applying L'Hôpital's rule repeatedly, $\frac{nx^{n-1}}{e^x} = \dots = \frac{n!}{e^x} \rightarrow 0$.

Proof of (3). By result (2), $\Gamma(n+1) = n\Gamma(n)$

$$\begin{aligned} \Rightarrow \Gamma(n+1) &= n(n-1)\Gamma(n-1) \quad (\because \text{by the same result, when } n \text{ is a positive integer}) \\ &= n(n-1)(n-2)\Gamma(n-2) = \dots = n(n-1)(n-2) \dots 2\Gamma(2) \end{aligned}$$

$$= n(n-1)(n-2) \cdots 2 \cdot 1 \Gamma(1) = n(n-1)(n-2) \cdots 2 \cdot 1 \quad (\because \Gamma(1) = 1)$$

$$\Rightarrow \boxed{\Gamma(n+1) = n!} \quad \text{where } n \text{ is a positive integer.}$$

Note

We have $\Gamma(n+1) = n\Gamma(n) \Rightarrow \Gamma(0) = \Gamma(-1+1) = (-1)!$, not defined. $\Rightarrow \Gamma(0)$ is an undefined one.

3 Beta integral

Definition 3.1 (Beta integral)

An improper integral of the form $\int_0^1 x^{m-1}(1-x)^{n-1} dx$, $m > 0$, $n > 0$, is called the **Beta integral** and is denoted by $\beta(m, n)$, i.e.

$$\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx.$$

Result 3.1 (Properties of the Beta integral)

(1) $\beta(m, n) = \beta(n, m)$, i.e. the Beta integral is symmetric about its parameters.

$$(2) \beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

$$(3) \beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

Proof of (1). By definition, $\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx$. Let $y = 1-x \Rightarrow x = 1-y \Rightarrow dx = -dy$. When $x = 0 \Rightarrow y = 1$; when $x = 1 \Rightarrow y = 0$.

$$\Rightarrow \beta(m, n) = \int_1^0 (1-y)^{m-1} y^{n-1} (-dy) = \int_0^1 y^{n-1} (1-y)^{m-1} dy = \beta(n, m) \quad (\because \text{by definition})$$

$$\Rightarrow \boxed{\beta(m, n) = \beta(n, m)}$$

Proof of (2). By definition, $\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx$ (1). Put $x = \sin^2 \theta \Rightarrow dx = 2 \sin \theta \cos \theta d\theta$. When $x = 0 \Rightarrow \theta = 0$; when $x = 1 \Rightarrow \theta = \frac{\pi}{2}$.

$$\therefore (1) \Rightarrow \beta(m, n) = \int_0^{\pi/2} (\sin^2 \theta)^{m-1} (1 - \sin^2 \theta)^{n-1} (2 \sin \theta \cos \theta) d\theta$$

$$= \int_0^{\pi/2} \sin^{2m-2} \theta \cos^{2n-2} \theta 2 \sin \theta \cos \theta d\theta \Rightarrow \boxed{\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta}$$

Proof of (3). By definition $\beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx$. Let

$$\boxed{x = \frac{y}{1+y}} \Rightarrow (1+y)x = y \Rightarrow y(x-1) = -x \Rightarrow \boxed{y = \frac{x}{1-x}}$$

When $x = 0 \Rightarrow y = 0$; when $x = 1 \Rightarrow y \rightarrow \infty$. Now

$$dx = \frac{(1+y) dy - y dy}{(1+y)^2} \Rightarrow dx = \frac{dy}{(1+y)^2}.$$

$$\therefore \beta(m, n) = \int_0^\infty \left(\frac{y}{1+y} \right)^{m-1} \left(1 - \frac{y}{1+y} \right)^{n-1} \frac{dy}{(1+y)^2}$$

$$\begin{aligned}
&= \int_0^\infty \frac{y^{m-1}}{(1+y)^{m-1}} \left(\frac{1}{1+y} \right)^{n-1} \frac{dy}{(1+y)^2} \\
&= \int_0^\infty \frac{y^{m-1}}{(1+y)^{m-1+n-1+2}} dy \Rightarrow \beta(m, n) = \int_0^\infty \frac{y^{m-1}}{(1+y)^{m+n}} dy.
\end{aligned}$$

Generally,

$$\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

Note

So far we have discussed: (1) $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$ ($n > 0$) and $\Gamma(n+1) = n\Gamma(n)$ (or $n!$); (2) $\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, $\beta(m, n) = \beta(n, m)$; (3) $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$; (4) $\beta(m, n) = \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}} dx$.

4 Relation between Beta and Gamma integrals

Result 4.1 (Beta–Gamma relation)

$$\beta(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

Proof. By definition, $\Gamma(m) = \int_0^\infty e^{-x} x^{m-1} dx$. Put $x = y^2 \Rightarrow dx = 2y dy$; when x varies from 0 to ∞ , y varies from 0 to ∞ .

$$\Rightarrow \Gamma(m) = \int_0^\infty e^{-y^2} y^{2m-2} 2y dy \Rightarrow \Gamma(m) = 2 \int_0^\infty e^{-y^2} y^{2m-1} dy.$$

In a similar way, $\Gamma(n) = 2 \int_0^\infty e^{-x^2} x^{2n-1} dx$.

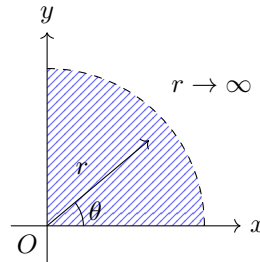
$$\Rightarrow \Gamma(m) \cdot \Gamma(n) = 2 \int_0^\infty e^{-y^2} y^{2m-1} dy \cdot 2 \int_0^\infty e^{-x^2} x^{2n-1} dx$$

$$\Gamma(m) \cdot \Gamma(n) = 4 \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2n-1} y^{2m-1} dx dy. \quad (1)$$

Let $x = r \cos \theta$ and $y = r \sin \theta$ ($(x, y) \rightarrow (r, \theta)$) $\Rightarrow x^2 + y^2 = r^2$. Under the double integration concept we can write

$$dx dy = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} dr d\theta = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} dr d\theta = (r \cos^2 \theta + r \sin^2 \theta) dr d\theta = r dr d\theta.$$

Under the transformation $x = r \cos \theta$ and $y = r \sin \theta$ we have the curve $x^2 + y^2 = r^2$, a circle. But here x varies from 0 to ∞ and y varies from 0 to ∞ , i.e. we have the region of the positive quadrant of a very large circle. In this region, r varies from 0 to ∞ and θ varies from 0 to $\frac{\pi}{2}$.



$$(1) \Rightarrow \Gamma(m) \Gamma(n) = 4 \int_{\theta=0}^{\pi/2} \int_{r=0}^{\infty} e^{-r^2} (r \cos \theta)^{2n-1} (r \sin \theta)^{2m-1} r dr d\theta$$

$$\begin{aligned}
&= \left(2 \int_0^\infty e^{-r^2} r^{2n-1} r^{2m-1} r dr \right) \left(2 \int_0^{\pi/2} \cos^{2n-1} \theta \sin^{2m-1} \theta d\theta \right) \\
&= \left(2 \int_0^\infty e^{-r^2} r^{2m+2n-1} dr \right) \left(2 \int_0^{\pi/2} \cos^{2n-1} \theta \sin^{2m-1} \theta d\theta \right) \\
&= \left(2 \int_0^\infty e^{-r^2} r^{2(m+n)-1} dr \right) \beta(m, n)
\end{aligned}$$

$$\Rightarrow \Gamma(m)\Gamma(n) = \Gamma(m+n) \beta(m, n) \quad \left(\because \Gamma(m) = 2 \int_0^\infty e^{-y^2} y^{2m-1} dy \right) \Rightarrow \boxed{\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}}$$

4.1 Consequences of the Beta–Gamma relation

Result (1). We know that $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$ and $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

$$\Rightarrow 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}.$$

Let $2m-1 = p$ and $2n-1 = q \Rightarrow m = \frac{p+1}{2}$ and $n = \frac{q+1}{2}$:

$$\Rightarrow 2 \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+1}{2} + \frac{q+1}{2}\right)}$$

Result 4.2 (Reduction formula)

$$\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)} \quad (1)$$

Result (2). In the above reduction formula (1), if we put $p = 0, q = 0$, then

$$\begin{aligned}
\int_0^{\pi/2} d\theta &= \frac{1}{2} \frac{\Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma(1)} \Rightarrow (\theta)_0^{\pi/2} = \frac{1}{2} \left(\Gamma\left(\frac{1}{2}\right)\right)^2 \quad (\because \Gamma(1) = 1) \Rightarrow \frac{1}{2} \left(\Gamma\left(\frac{1}{2}\right)\right)^2 = \frac{\pi}{2} \\
&\Rightarrow \left(\Gamma\left(\frac{1}{2}\right)\right)^2 = \pi \Rightarrow \boxed{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}
\end{aligned} \quad (2)$$

Result (3). In the above reduction formula (1), if we put $q = 0$, then

$$\int_0^{\pi/2} \sin^p \theta d\theta = \frac{1}{2} \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{p+2}{2}\right)} \Rightarrow \boxed{\int_0^{\pi/2} \sin^p \theta d\theta = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p+2}{2}\right)}} \quad (3)$$

Result (4). Similarly, if we put $p = 0$ in (1), then

$$\boxed{\int_0^{\pi/2} \cos^q \theta d\theta = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{q+2}{2}\right)}} \quad (4)$$

Note

Remark. We know $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$ and $\Gamma(n+1) = n\Gamma(n)$. Suppose we wish to identify

$$\Gamma\left(\frac{3}{2}\right) = \Gamma\left(\frac{1}{2} + 1\right) = \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{1}{2} \sqrt{\pi} \Rightarrow \boxed{\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}};$$

similarly $\Gamma\left(\frac{5}{2}\right) = \Gamma\left(\frac{3}{2} + 1\right) = \frac{3}{2} \Gamma\left(\frac{3}{2}\right) = \frac{3}{2} \left(\frac{\sqrt{\pi}}{2}\right) \Rightarrow \boxed{\Gamma\left(\frac{5}{2}\right) = \frac{3\sqrt{\pi}}{4}}$. Similarly we can get the values for

$\Gamma\left(\frac{7}{2}\right), \Gamma\left(\frac{9}{2}\right), \Gamma\left(\frac{11}{2}\right)$, etc.

Suppose we wish to identify Γ at negative fractions. Using $\Gamma(n) = \frac{\Gamma(n+1)}{n}$: for $n = -\frac{1}{2}$,

$$\Gamma\left(-\frac{1}{2}\right) = \frac{\Gamma\left(-\frac{1}{2} + 1\right)}{-\frac{1}{2}} = -2\Gamma\left(\frac{1}{2}\right) \Rightarrow \boxed{\Gamma\left(-\frac{1}{2}\right) = -2\sqrt{\pi}}$$

For $n = -\frac{3}{2}$:

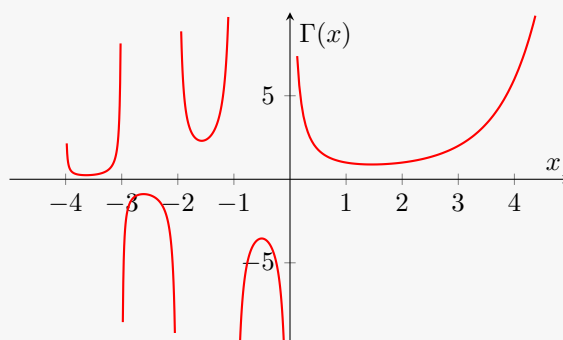
$$\Gamma\left(-\frac{3}{2}\right) = \frac{\Gamma\left(-\frac{3}{2} + 1\right)}{-\frac{3}{2}} = -\frac{2}{3}\Gamma\left(-\frac{1}{2}\right) = -\frac{2}{3}(-2\sqrt{\pi}) \Rightarrow \boxed{\Gamma\left(-\frac{3}{2}\right) = \frac{4}{3}\sqrt{\pi}}.$$

Similarly we can derive for $n = -\frac{5}{2}, -\frac{7}{2}, \dots$

Note

From these results we see that $\Gamma(n)$ is defined for $n > 0$ (by definition), and $\Gamma(n)$ is defined for negative fractional values except negative integers. $\therefore$ We can use the definition of $\Gamma(n)$ as follows:

$$\Gamma(n) = \begin{cases} \int_0^\infty e^{-x} x^{n-1} dx, & n > 0 \\ \frac{\Gamma(n+1)}{n}, & -1 < n < 0, -2 < n < -1, \dots \end{cases}$$



4.2 Duplication formula for $\Gamma(n)$

Result 4.3 (Duplication formula)

$$\Gamma(n) \Gamma\left(n + \frac{1}{2}\right) = \frac{1}{2^{2n-1}} \sqrt{\pi} \Gamma(2n)$$

Proof. We know $\beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$

$$\begin{aligned} \Rightarrow \beta(n, n) &= 2 \int_0^{\pi/2} \sin^{2n-1} \theta \cos^{2n-1} \theta d\theta = 2 \int_0^{\pi/2} (\sin \theta \cos \theta)^{2n-1} d\theta \\ &= 2 \int_0^{\pi/2} \left(\frac{\sin 2\theta}{2}\right)^{2n-1} d\theta = \frac{2}{2^{2n-1}} \int_0^{\pi/2} \sin^{2n-1} 2\theta d\theta. \end{aligned}$$

Let $2\theta = \alpha \Rightarrow \theta = \frac{\alpha}{2} \Rightarrow d\theta = \frac{d\alpha}{2}$. When $\theta = 0 \Rightarrow \alpha = 0$; $\theta = \frac{\pi}{2} \Rightarrow \alpha = \pi$.

$$= \frac{2}{2^{2n-1}} \int_0^\pi \sin^{2n-1} \alpha \frac{d\alpha}{2} = \frac{1}{2^{2n-1}} \int_0^\pi \sin^{2n-1} \alpha d\alpha. \quad (1)$$

We use $\int_0^{2a} f(x) dx = 2 \int_0^a f(x) dx$ if $f(2a-x) = f(x)$. Let $2a = \pi \Rightarrow a = \frac{\pi}{2}$ and $f(\alpha) = \sin^{2n-1} \alpha = (\sin \alpha)^{2n-1}$.

Then $f(2a - \alpha) = f(\pi - \alpha) = (\sin(\pi - \alpha))^{2n-1} = (\sin \alpha)^{2n-1} = f(\alpha)$, so $\int_0^\pi f(\alpha) d\alpha = 2 \int_0^{\pi/2} f(\alpha) d\alpha$.

$$\begin{aligned} \therefore (1) \Rightarrow \beta(n, n) &= \frac{1}{2^{2n-1}} 2 \int_0^{\pi/2} \sin^{2n-1} \alpha d\alpha = \frac{1}{2^{2n-1}} \beta\left(n, \frac{1}{2}\right) \\ &\left(\because \beta(m, n) = 2 \int_0^{\pi/2} \sin^{2m-1} \alpha \cos^{2n-1} \alpha d\alpha, n = \frac{1}{2} \right) \\ \Rightarrow \frac{\Gamma(n)\Gamma(n)}{\Gamma(n+n)} &= \frac{1}{2^{2n-1}} \frac{\Gamma(n)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(n + \frac{1}{2}\right)} \Rightarrow \frac{\Gamma(n)}{\Gamma(2n)} = \frac{1}{2^{2n-1}} \frac{\sqrt{\pi}}{\Gamma\left(n + \frac{1}{2}\right)} \\ \Rightarrow \Gamma(n) \Gamma\left(n + \frac{1}{2}\right) &= \frac{1}{2^{2n-1}} \sqrt{\pi} \Gamma(2n) \end{aligned}$$

Note

Remark. By the duplication formula, $\Gamma(n)\Gamma\left(n + \frac{1}{2}\right) = \frac{1}{2^{2n-1}} \sqrt{\pi} \Gamma(2n)$.

Put $n = \frac{1}{4}$: $\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right) = \frac{1}{2^{-1/2}} \sqrt{\pi} \Gamma\left(\frac{1}{2}\right) = \sqrt{2} \sqrt{\pi} \sqrt{\pi} \Rightarrow \boxed{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{3}{4}\right) = \pi \sqrt{2}}$.

Put $n = \frac{3}{4}$:

$$\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{5}{4}\right) = \frac{1}{2^{1/2}} \sqrt{\pi} \Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{\sqrt{2}} \Gamma\left(\frac{1}{2} + 1\right) = \sqrt{\frac{\pi}{2}} \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \sqrt{\frac{\pi}{2}} \frac{1}{2} \sqrt{\pi} \Rightarrow \boxed{\Gamma\left(\frac{3}{4}\right)\Gamma\left(\frac{5}{4}\right) = \frac{\pi}{2\sqrt{2}}}$$

Similarly we can get the values for $\Gamma\left(\frac{5}{4}\right)\Gamma\left(\frac{7}{4}\right)$, $\Gamma\left(\frac{7}{4}\right)\Gamma\left(\frac{9}{4}\right)$, and so on.

5 Problems on Beta and Gamma integrals

Problem 5.1

Prove that $\int_0^\infty e^{-ax} x^{n-1} dx = \frac{\Gamma(n)}{a^n}$, $n > 0$.

Solution. Let $I = \int_0^\infty e^{-ax} x^{n-1} dx$. Let $y = ax \Rightarrow x = \frac{y}{a} \Rightarrow dx = \frac{dy}{a}$; when x varies from 0 to ∞ , y varies from 0 to ∞ .

$$\Rightarrow I = \int_0^\infty e^{-y} \left(\frac{y}{a}\right)^{n-1} \frac{dy}{a} = \frac{1}{a^n} \int_0^\infty e^{-y} y^{n-1} dy \Rightarrow \boxed{I = \frac{\Gamma(n)}{a^n}} \quad (\because \text{by definition})$$

■

Problem 5.2

Evaluate $\int_0^{\pi/2} \sin^8 \theta d\theta$ and $\int_0^{\pi/2} \cos^5 \theta d\theta$.

Solution. (i) Given $\int_0^{\pi/2} \sin^8 \theta d\theta = \int_0^{\pi/2} \sin^p \theta d\theta$ where $p = 8$

$$\begin{aligned} &= \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{p+1}{2}\right)}{\Gamma\left(\frac{p+2}{2}\right)} = \frac{\sqrt{\pi}}{2} \frac{\Gamma\left(\frac{9}{2}\right)}{\Gamma(5)} \\ &= \frac{\sqrt{\pi}}{2} \frac{\frac{7}{2} \cdot \frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}}{4!} \quad (\because \Gamma(n) = (n-1)!) \\ &= \frac{\sqrt{\pi}}{2} \frac{105\sqrt{\pi}}{16 \times 24} = \frac{105\pi}{32 \times 24} \Rightarrow \boxed{\int_0^{\pi/2} \sin^8 \theta d\theta = \frac{105\pi}{768}} \quad \left(= \frac{35\pi}{256}\right) \end{aligned}$$

(ii) $\int_0^{\pi/2} \cos^5 \theta d\theta = \int_0^{\pi/2} \cos^q \theta d\theta$ with $q = 5$

$$= \frac{\sqrt{\pi} \Gamma\left(\frac{q+1}{2}\right)}{2 \Gamma\left(\frac{q+2}{2}\right)} = \frac{\sqrt{\pi} \Gamma(3)}{2 \Gamma\left(\frac{7}{2}\right)} = \frac{\sqrt{\pi}}{2} \frac{2!}{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}} = \frac{\sqrt{\pi}}{2} \cdot \frac{2 \times 8}{15\sqrt{\pi}} \Rightarrow \boxed{\int_0^{\pi/2} \cos^5 \theta d\theta = \frac{8}{15}}$$

■

Problem 5.3

Evaluate $\int_0^{\pi/2} \sin^2 \theta \cos^3 \theta d\theta$.

Solution. Given $\int_0^{\pi/2} \sin^2 \theta \cos^3 \theta d\theta = \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta$ where $p = 2, q = 3$

$$\begin{aligned} &= \frac{1}{2} \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{\Gamma\left(\frac{p+q+2}{2}\right)} \\ &= \frac{1}{2} \frac{\Gamma\left(\frac{3}{2}\right) \Gamma\left(\frac{4}{2}\right)}{\Gamma\left(\frac{7}{2}\right)} \\ &= \frac{1}{2} \frac{\frac{1}{2} \sqrt{\pi} (1!)}{\frac{5}{2} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}} = \frac{1}{2} \cdot \frac{4}{15} \\ &\Rightarrow \boxed{\int_0^{\pi/2} \sin^2 \theta \cos^3 \theta d\theta = \frac{4}{30} = \frac{2}{15}} \end{aligned}$$

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Problem 5.4

Evaluate $\int_0^{\infty} e^{-2x} x^6 dx$.

Solution. Given $\int_0^{\infty} e^{-2x} x^6 dx = \int_0^{\infty} e^{-ax} x^{n-1} dx$ where $a = 2, n = 7$

$$= \frac{\Gamma(n)}{a^n} = \frac{\Gamma(7)}{2^7} = \frac{6!}{128} = \frac{720}{128} = \frac{360}{64} = \frac{180}{32} = \boxed{\frac{45}{8}}$$

■

Problem 5.5

Evaluate $\int_0^1 (x \log_e x)^3 dx$.

Solution. Given $I = \int_0^1 (x \log_e x)^3 dx$. Let $x = e^{-t} \Rightarrow \log_e x = -t$ and $dx = -e^{-t} dt$. When $x = 0 \Rightarrow e^{-t} = 0 \Rightarrow t \rightarrow \infty$ ($\because e^{-\infty} = 0$); when $x = 1 \Rightarrow e^{-t} = 1 \Rightarrow t = 0$.

$$\begin{aligned} \therefore I &= \int_{\infty}^0 (e^{-t}(-t))^3 (-e^{-t} dt) = \int_{\infty}^0 e^{-4t} t^3 dt = - \int_0^{\infty} e^{-4t} t^3 dt \\ &= - \int_0^{\infty} e^{-4t} t^{4-1} dt = - \frac{\Gamma(4)}{4^4} = - \frac{6}{256} \Rightarrow \boxed{I = - \frac{3}{128}} \end{aligned}$$

■

Problem 5.6

Evaluate $\int_0^1 x^3 (1-x)^4 dx$.

Solution. Given $\int_0^1 x^3(1-x)^4 dx = \int_0^1 x^{m-1}(1-x)^{n-1} dx$ where $m = 4, n = 5$

$$= \beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)} = \frac{\Gamma(4)\Gamma(5)}{\Gamma(9)} = \frac{3!4!}{8!} = \frac{6 \times 24}{40320} \Rightarrow \boxed{\int_0^1 x^3(1-x)^4 dx = \frac{1}{280}}$$

■

Problem 5.7

Derive the value of $\int_0^1 x^m(1-x^n)^p dx$.

Solution. Given $I = \int_0^1 x^m(1-x^n)^p dx$. Let $y = x^n \Rightarrow y^{1/n} = x \Rightarrow dx = \frac{1}{n}y^{\frac{1}{n}-1}dy$. When x varies from 0 to 1, y varies from 0 to 1.

$$\begin{aligned} \Rightarrow I &= \int_0^1 \left(y^{1/n}\right)^m (1-y)^p \frac{1}{n} y^{\frac{1}{n}-1} dy = \frac{1}{n} \int_0^1 y^{\frac{m}{n}+\frac{1}{n}-1} (1-y)^p dy \\ &= \frac{1}{n} \int_0^1 y^{\frac{m+1}{n}-1} (1-y)^{(p+1)-1} dy = \frac{1}{n} \beta\left(\frac{m+1}{n}, p+1\right) \quad \left(\because \beta(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1} dx\right) \\ &\Rightarrow \boxed{\int_0^1 x^m(1-x^n)^p dx = \frac{1}{n} \beta\left(\frac{m+1}{n}, p+1\right) = \frac{1}{n} \frac{\Gamma\left(\frac{m+1}{n}\right) \Gamma(p+1)}{\Gamma\left(\frac{m+1}{n} + p+1\right)}} \end{aligned}$$

■

Problem 5.8

Evaluate (i) $\int_0^1 x^2(1-x^3)^4 dx$ and (ii) $\int_0^1 \frac{dx}{\sqrt{1-x^n}}$.

Solution. Already we proved that $\int_0^1 x^m(1-x^n)^p dx = \frac{1}{n} \beta\left(\frac{m+1}{n}, p+1\right)$.

(i) Let $I = \int_0^1 x^2(1-x^3)^4 dx = \int_0^1 x^m(1-x^n)^p dx$ where $m = 2, n = 3$ and $p = 4$

$$= \frac{1}{n} \beta\left(\frac{m+1}{n}, p+1\right) = \frac{1}{3} \beta\left(\frac{3}{3}, 5\right) = \frac{1}{3} \beta(1, 5) = \frac{1}{3} \frac{\Gamma(1)\Gamma(5)}{\Gamma(6)} = \frac{1}{3} \cdot \frac{1 \cdot 4!}{5!} = \frac{1}{3} \cdot \frac{1}{5} \Rightarrow \boxed{I = \frac{1}{15}}$$

(ii) Given

$$\begin{aligned} I &= \int_0^1 \frac{dx}{\sqrt{1-x^n}} = \int_0^1 (1-x^n)^{-1/2} dx = \int_0^1 x^m(1-x^n)^p dx, \quad m = 0, n = n, p = -\frac{1}{2} \\ &= \frac{1}{n} \beta\left(\frac{m+1}{n}, p+1\right) = \frac{1}{n} \beta\left(\frac{1}{n}, \frac{1}{2}\right) = \frac{1}{n} \frac{\Gamma\left(\frac{1}{n}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{n} + \frac{1}{2}\right)} \Rightarrow \boxed{I = \frac{\sqrt{\pi}}{n} \frac{\Gamma\left(\frac{1}{n}\right)}{\Gamma\left(\frac{2+n}{2n}\right)}} \end{aligned}$$

■

Problem 5.9

Prove that $\int_0^1 \frac{x^2 dx}{\sqrt{1-x^4}} \times \int_0^1 \frac{dx}{\sqrt{1+x^4}} = \frac{\pi}{4\sqrt{2}}$.

Solution. Let $I = I_1 \times I_2$, where $I_1 = \int_0^1 \frac{x^2}{\sqrt{1-x^4}} dx$ and $I_2 = \int_0^1 \frac{dx}{\sqrt{1+x^4}}$.

Now $I_1 = \int_0^1 x^2(1-x^4)^{-1/2} dx = \int_0^1 x^m(1-x^n)^p dx$ where $m = 2, n = 4, p = -\frac{1}{2}$

$$= \frac{1}{n} \beta\left(\frac{m+1}{n}, p+1\right) \Rightarrow I_1 = \frac{1}{4} \beta\left(\frac{3}{4}, \frac{1}{2}\right) = \frac{1}{4} \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{4} + \frac{1}{2}\right)} \Rightarrow \boxed{I_1 = \frac{1}{4} \frac{\Gamma\left(\frac{3}{4}\right) \sqrt{\pi}}{\Gamma\left(\frac{5}{4}\right)}}$$

Now $I_2 = \int_0^1 \frac{dx}{\sqrt{1+x^4}}$. Put $x^2 = \tan \theta \Rightarrow 2x dx = \sec^2 \theta d\theta \Rightarrow dx = \frac{\sec^2 \theta}{2x} d\theta = \frac{\sec^2 \theta}{2\sqrt{\tan \theta}} d\theta$. When $x = 0 \Rightarrow \theta = 0$; $x = 1 \Rightarrow \theta = \frac{\pi}{4}$.

$$\begin{aligned} \therefore I_2 &= \int_0^{\pi/4} \frac{1}{\sqrt{1+\tan^2 \theta}} \frac{\sec^2 \theta}{2\sqrt{\tan \theta}} d\theta \\ &= \frac{1}{2} \int_0^{\pi/4} \frac{1}{\sec \theta} \frac{\sec^2 \theta}{\sqrt{\tan \theta}} d\theta \\ &= \frac{1}{2} \int_0^{\pi/4} \frac{\sec \theta}{\sqrt{\tan \theta}} d\theta \\ &= \frac{1}{2} \int_0^{\pi/4} \frac{1}{\cos \theta} \times \frac{\cos^{1/2} \theta}{\sin^{1/2} \theta} d\theta \\ &= \frac{1}{2} \int_0^{\pi/4} \sin^{-1/2} \theta \cos^{-1/2} \theta d\theta \end{aligned}$$

$\Rightarrow I_2 = \frac{1}{2} \int_0^a f(\theta) d\theta$ where $a = \frac{\pi}{4}$, $f(\theta) = (\sin \theta \cos \theta)^{-1/2}$. Then $f(2a-\theta) = f(\frac{\pi}{2} - \theta) = (\sin(\frac{\pi}{2} - \theta) \cos(\frac{\pi}{2} - \theta))^{-1/2} = (\cos \theta \sin \theta)^{-1/2} = f(\theta) \Rightarrow \int_0^{2a} f(\theta) d\theta = 2 \int_0^a f(\theta) d\theta$.

$$\begin{aligned} \Rightarrow I_2 &= \frac{1}{2} \left(\frac{1}{2} \int_0^{2a} f(\theta) d\theta \right) = \frac{1}{4} \int_0^{\pi/2} \sin^{-1/2} \theta \cos^{-1/2} \theta d\theta \\ &= \frac{1}{4} \left(\frac{1}{2} \frac{\Gamma(\frac{1}{4}) \Gamma(\frac{1}{4})}{\Gamma(\frac{1}{2})} \right) \left(\because \int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} \frac{\Gamma(\frac{p+1}{2}) \Gamma(\frac{q+1}{2})}{\Gamma(\frac{p+q+2}{2})} \right) \\ &\Rightarrow \boxed{I_2 = \frac{1}{8} \frac{(\Gamma(\frac{1}{4}))^2}{\sqrt{\pi}}} \end{aligned}$$

$\therefore I = I_1 \times I_2$:

$$\begin{aligned} I &= \frac{1}{4} \frac{\sqrt{\pi} \Gamma(\frac{3}{4})}{\Gamma(\frac{5}{4})} \times \frac{1}{8} \frac{(\Gamma(\frac{1}{4}))^2}{\sqrt{\pi}} \\ &= \frac{1}{32} \frac{\Gamma(\frac{3}{4}) (\Gamma(\frac{1}{4}))^2}{\Gamma(\frac{1}{4} + 1)} \\ &= \frac{1}{32} \frac{\Gamma(\frac{3}{4}) (\Gamma(\frac{1}{4}))^2}{\frac{1}{4} \Gamma(\frac{1}{4})} \quad (\because \Gamma(n+1) = n\Gamma(n)) \\ &= \frac{1}{8} \Gamma(\frac{1}{4}) \Gamma(\frac{3}{4}) = \frac{1}{8} \pi \sqrt{2} \quad (\because \text{by the duplication formula}) \Rightarrow \boxed{I = \frac{\pi}{4\sqrt{2}}} \end{aligned}$$

■

Problem 5.10

Prove that $\int_0^\infty \frac{e^{-x^2}}{\sqrt{x}} dx \times \int_0^\infty x^2 e^{-x^4} dx = \frac{\pi}{4\sqrt{2}}$.

Solution. Let $I = I_1 \times I_2$, where $I_1 = \int_0^\infty e^{-x^2} x^{-1/2} dx$ and $I_2 = \int_0^\infty e^{-x^4} x^2 dx$. Recall $\Gamma(n) = \int_0^\infty e^{-x} x^{n-1} dx$.

Now, in I_1 let $y = x^2 \Rightarrow 2x dx = dy \Rightarrow dx = \frac{dy}{2x} = \frac{dy}{2\sqrt{y}}$; there is no change in the limits.

$$\begin{aligned} \Rightarrow I_1 &= \int_0^\infty e^{-y} (y^{1/2})^{-1/2} \frac{dy}{2\sqrt{y}} = \frac{1}{2} \int_0^\infty e^{-y} y^{-\frac{1}{4}-\frac{1}{2}} dy \\ &= \frac{1}{2} \int_0^\infty e^{-y} y^{-3/4} dy = \frac{1}{2} \int_0^\infty e^{-y} y^{\frac{1}{4}-1} dy \Rightarrow \boxed{I_1 = \frac{1}{2} \Gamma(\frac{1}{4})} \end{aligned}$$

Similarly, in $I_2 = \int_0^\infty e^{-x^4} x^2 dx$ let $y = x^4 \Rightarrow x = y^{1/4}$, $x^2 = y^{1/2}$, $dx = \frac{1}{4}y^{-3/4}dy$:

$$I_2 = \frac{1}{4} \int_0^\infty e^{-y} y^{\frac{1}{2}-\frac{3}{4}} dy = \frac{1}{4} \int_0^\infty e^{-y} y^{\frac{3}{4}-1} dy \Rightarrow \boxed{I_2 = \frac{1}{4} \Gamma\left(\frac{3}{4}\right)}$$

$$\therefore I = I_1 \times I_2 = \frac{1}{2} \Gamma\left(\frac{1}{4}\right) \times \frac{1}{4} \Gamma\left(\frac{3}{4}\right) = \frac{1}{8} \Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{3}{4}\right) = \frac{1}{8} (\pi\sqrt{2}) \quad (\because \text{by the duplication formula}) \Rightarrow \boxed{I = \frac{\pi}{4\sqrt{2}}}. \quad \blacksquare$$

Problem 5.11

Evaluate (i) $\int_a^b (x-a)^{m-1} (b-x)^{n-1} dx$ (ii) $\int_{-a}^a (a+x)^{m-1} (a-x)^{n-1} dx$ in terms of the Beta integral.

Solution. (i) Let $I = \int_a^b (x-a)^{m-1} (b-x)^{n-1} dx$. Recall $\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$. Let $y = x-a \Rightarrow x = y+a \Rightarrow dx = dy$; when $x=a \Rightarrow y=0$; when $x=b \Rightarrow y=b-a$.

$$\begin{aligned} \therefore I &= \int_0^{b-a} y^{m-1} (b-(y+a))^{n-1} dy = \int_0^{b-a} y^{m-1} ((b-a)-y)^{n-1} dy \\ &= \int_0^{b-a} y^{m-1} \left((b-a) \left(1 - \frac{y}{b-a} \right) \right)^{n-1} dy = (b-a)^{n-1} \int_0^{b-a} y^{m-1} \left(1 - \frac{y}{b-a} \right)^{n-1} dy. \end{aligned}$$

Let $z = \frac{y}{b-a} \Rightarrow y = (b-a)z$, $dy = (b-a) dz$; when $y=0 \Rightarrow z=0$; when $y=b-a \Rightarrow z=1$.

$$\begin{aligned} \Rightarrow I &= (b-a)^{n-1} \int_0^1 ((b-a)z)^{m-1} (1-z)^{n-1} (b-a) dz \\ &= (b-a)^{n-1+m-1+1} \int_0^1 z^{m-1} (1-z)^{n-1} dz \\ &\Rightarrow \boxed{I = (b-a)^{m+n-1} \beta(m, n)} \end{aligned}$$

(ii) Similarly, with $b=a$ and $a \rightarrow -a$ in (i) (i.e. $x-(-a)=a+x$ and $b-x=a-x$):

$$\int_{-a}^a (a+x)^{m-1} (a-x)^{n-1} dx = (2a)^{m+n-1} \beta(m, n). \quad \blacksquare$$

6 Multiple integrals

This concept involves the evaluation of integrals of functions w.r.t. x and y , and similarly w.r.t. x , y and z , called **double integration** and **triple integration** respectively.

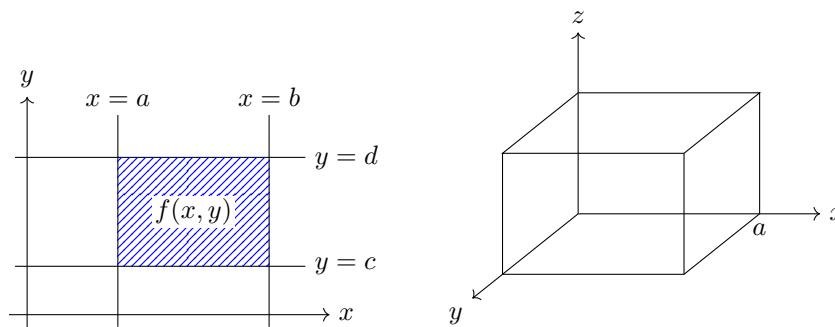
Definition 6.1 (Double and triple integrals)

A **double integral** is generally denoted by

$$\int_{y=c}^d \int_{x=a}^b f(x, y) dx dy,$$

and a **triple integral** is usually denoted by

$$\int_{z=0}^c \int_{y=0}^b \int_{x=0}^a f(x, y, z) dx dy dz.$$

**Note**

Remark. Evaluation of double and triple integration takes some order based on the region we consider and also the limit conditions.

6.1 Evaluation of double integrals

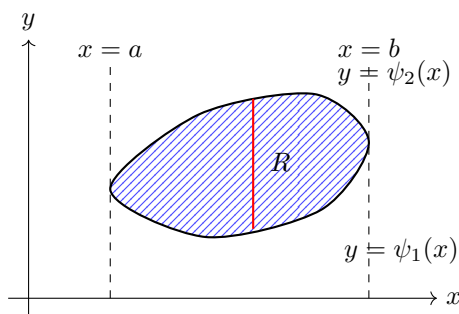
Case (1): Rectangular region. Suppose we have the square (or rectangular) region R , $a \leq x \leq b$, $c \leq y \leq d$, where a, b, c, d are constants. Then

$$\iint_R f(x, y) dy dx = \int_{x=a}^b \left(\int_{y=c}^d f(x, y) dy \right) dx \quad (\text{integrate partially w.r.t. } y \text{ and keep } x \text{ as constant temporarily})$$

$$(\text{OR}) \quad = \int_{y=c}^d \left(\int_{x=a}^b f(x, y) dx \right) dy \quad (\text{same procedure}).$$

Case (2): Curved region. Suppose we have the curved region R bounded by $x = \phi_1(y)$, $x = \phi_2(y)$, $y = c$, $y = d$ (or by $y = \psi_1(x)$, $y = \psi_2(x)$, $x = a$, $x = b$). Then

$$\iint_R f(x, y) dx dy = \int_{y=c}^d \left(\int_{x=\phi_1(y)}^{\phi_2(y)} f(x, y) dx \right) dy \quad (\text{OR}) \quad = \int_{x=a}^b \left(\int_{y=\psi_1(x)}^{\psi_2(x)} f(x, y) dy \right) dx.$$

**Result 6.1 (Rule for the limits)**

The **inner** integral has the **variable** limits, and the **outer** integral has the **constant** limits.

6.2 Problems

Problem 6.1

Evaluate $\int_0^2 \int_0^1 4xy dx dy$.

Solution. Given

$$I = \int_{y=0}^2 \left(\int_{x=0}^1 4xy dx \right) dy = \int_0^2 4y \left(\frac{x^2}{2} \right)_0^1 dy = \int_0^2 2y(1) dy$$

$$= 2 \int_0^2 y \, dy = 2 \left(\frac{y^2}{2} \right)_0^2 = 4 \Rightarrow \boxed{I = 4}$$

(OR) by change of order:

$$I = \int_{x=0}^1 \left(\int_{y=0}^2 4xy \, dy \right) dx = \int_0^1 4x \left(\frac{y^2}{2} \right)_0^2 dx = \int_0^1 2x(4) \, dx = \int_0^1 8x \, dx = 8 \left(\frac{x^2}{2} \right)_0^1 = 4(1-0) \Rightarrow \boxed{I = 4}$$

($\because$ both limits are constants, the change of order is simply done by changing the limits only.) ■

Problem 6.2

Evaluate $\int_0^{\pi/2} \int_0^{\pi/2} \sin(\theta + \phi) \, d\theta \, d\phi$.

Solution. Let

$$\begin{aligned} I &= \int_{\phi=0}^{\pi/2} \left(\int_{\theta=0}^{\pi/2} \sin(\theta + \phi) \, d\theta \right) d\phi = \int_0^{\pi/2} (-\cos(\theta + \phi))_0^{\pi/2} d\phi \\ &= - \int_0^{\pi/2} [\cos(\frac{\pi}{2} + \phi) - \cos(0 + \phi)] \, d\phi = - \int_0^{\pi/2} (-\sin \phi - \cos \phi) \, d\phi \\ &= \int_0^{\pi/2} (\sin \phi + \cos \phi) \, d\phi = -(\cos \phi)_0^{\pi/2} + (\sin \phi)_0^{\pi/2} = -(0-1) + (1-0) = 1+1 \Rightarrow \boxed{I = 2} \end{aligned}$$

■

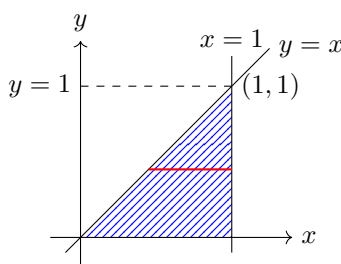
Problem 6.3

Evaluate $\int_0^1 \int_0^x dx \, dy$.

Solution. Let $I = \int_0^1 \int_0^x dx \, dy$ (wrong order, since the inner limit depends on x)

$$= \int_{x=0}^1 \left(\int_{y=0}^x dy \right) dx \quad (\text{rectified}) = \int_0^1 (y)_0^x dx = \int_0^1 x \, dx = \left(\frac{x^2}{2} \right)_0^1 \Rightarrow \boxed{I = \frac{1}{2}}$$

Aliter. In this problem, we have x varies from 0 to 1 and y varies from $y = 0$ to $y = x$; i.e. we have the following graph and region.

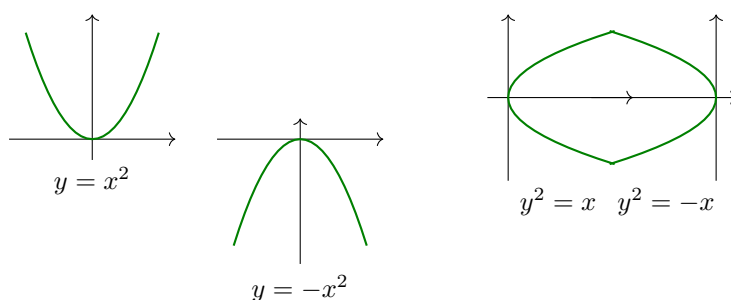


If we want to change the order of integration, i.e. $dy \, dx$ to $dx \, dy$, draw a horizontal strip in that region; it says that x varies from $x = y$ to $x = 1$ and y varies from $y = 0$ to $y = 1$.

$$\begin{aligned} \Rightarrow \int_{x=0}^1 \int_{y=0}^x dy \, dx &= \int_{y=0}^1 \left(\int_{x=y}^1 dx \right) dy \quad (\because \text{by change of order}) \\ &= \int_0^1 (x)_y^1 dy = \int_0^1 (1-y) \, dy = (y)_0^1 - \left(\frac{y^2}{2} \right)_0^1 = 1 - \frac{1}{2} = \frac{1}{2} \Rightarrow \boxed{I = \frac{1}{2}} \end{aligned}$$

■

6.3 Equations of standard curves



Straight lines. The intercept form is $\frac{x}{a} + \frac{y}{b} = 1$ (passing through $(a, 0)$ and $(0, b)$). E.g. $2x + 3y = 1 \Rightarrow \frac{x}{1/2} + \frac{y}{1/3} = 1$; $y = 2 - x \Rightarrow x + y = 2 \Rightarrow \frac{x}{2} + \frac{y}{2} = 1$. The line $y = mx$ passes through the origin.

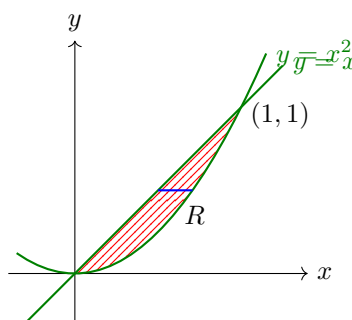
Circles. $x^2 + y^2 = a^2$: centre $(0, 0)$, radius a (so $y = \pm\sqrt{a^2 - x^2}$); $(x - a)^2 + (y - b)^2 = c^2$: centre (a, b) , radius c .

Hyperbola $xy = c^2$; **Ellipse** $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (through $(\pm a, 0)$ and $(0, \pm b)$).

Problem 6.4

Evaluate $\iint_R xy \, dx \, dy$, where R is the region bounded by the curves $y = x$ and $y = x^2$.

Solution. Let $I = \iint_R xy \, dx \, dy$, where R is the region bounded by the curves $y = x$ and $y = x^2$. $\therefore$ We have the following curves and region. Points of intersection: $y = x$ and $y = x^2 \Rightarrow x^2 - x = 0 \Rightarrow x(x - 1) = 0 \Rightarrow \boxed{x = 0, 1}$, i.e. $(0, 0)$ and $(1, 1)$.



In this region R , if we draw a horizontal strip, we have x varies from $y = x$ to $y = x^2$, i.e. x varies from $\boxed{x = y \text{ to } x = \sqrt{y}}$ and y varies from $\boxed{y = 0 \text{ to } y = 1}$. Then

$$\begin{aligned} I &= \iint_R xy \, dx \, dy = \int_{y=0}^1 \left(\int_{x=y}^{\sqrt{y}} xy \, dx \right) dy = \int_0^1 y \left(\frac{x^2}{2} \right)_y^{\sqrt{y}} dy \\ &= \frac{1}{2} \int_0^1 y(y - y^2) dy = \frac{1}{2} \int_0^1 (y^2 - y^3) dy = \frac{1}{2} \left[\left(\frac{y^3}{3} \right)_0^1 - \left(\frac{y^4}{4} \right)_0^1 \right] \\ &= \frac{1}{2} \left[\frac{1}{3} - \frac{1}{4} \right] = \frac{1}{2} \left(\frac{1}{12} \right) \Rightarrow \boxed{I = \frac{1}{24}} \end{aligned}$$

Aliter. Suppose in the above problem, in the region R , if we draw vertical strips, we have y varies from $y = x^2$ to $y = x$, i.e. $\boxed{y = x^2 \text{ to } y = x}$, and x varies from $\boxed{x = 0 \text{ to } 1}$.

$$\begin{aligned} \therefore I &= \iint_R xy \, dx \, dy = \int_{x=0}^1 \left(\int_{y=x^2}^x xy \, dy \right) dx = \int_0^1 x \left(\frac{y^2}{2} \right)_{x^2}^x dx \\ &= \frac{1}{2} \int_0^1 x(x^2 - x^4) dx = \frac{1}{2} \int_0^1 (x^3 - x^5) dx \end{aligned}$$

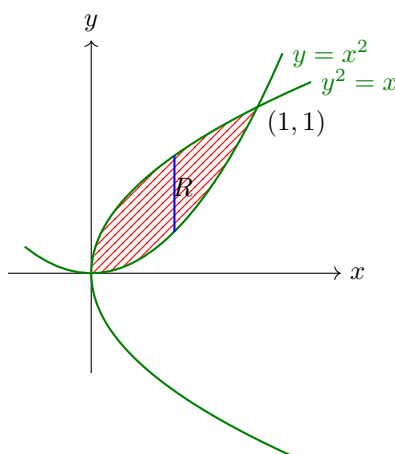
$$= \frac{1}{2} \left[\left(\frac{x^4}{4} \right)_0^1 - \left(\frac{x^6}{6} \right)_0^1 \right] = \frac{1}{2} \left(\frac{1}{4} - \frac{1}{6} \right) = \frac{1}{2} \left(\frac{3-2}{12} \right) = \frac{1}{24} \Rightarrow \boxed{I = \frac{1}{24}}$$

■

Problem 6.5

Evaluate $\iint_R xy \, dx \, dy$, where R is the region bounded by the two parabolas $y = x^2$ and $y^2 = x$.

Solution. Let $I = \iint_R xy \, dx \, dy$ where R is the region bounded by the parabolas $y = x^2$ and $y^2 = x$ (intersecting at $(0, 0)$ and $(1, 1)$).



In this region R , if we draw a vertical strip, y varies from $y = x^2$ to $y^2 = x$, i.e. y varies from $\boxed{y = x^2 \text{ to } \sqrt{x}}$ and x varies from $\boxed{x = 0 \text{ to } 1}$.

$$\begin{aligned} \text{Then } I &= \iint_R xy \, dx \, dy = \int_{x=0}^1 \left(\int_{y=x^2}^{\sqrt{x}} xy \, dy \right) dx = \int_0^1 x \left(\frac{y^2}{2} \right)_{x^2}^{\sqrt{x}} dx \\ &= \frac{1}{2} \int_0^1 x(x - x^4) dx \\ &= \frac{1}{2} \int_0^1 (x^2 - x^5) dx = \frac{1}{2} \left[\left(\frac{x^3}{3} \right)_0^1 - \left(\frac{x^6}{6} \right)_0^1 \right] = \frac{1}{2} \left[\frac{1}{3} - \frac{1}{6} \right] \Rightarrow \boxed{I = \frac{1}{12}} \end{aligned}$$

Exercise. Give the limits and check the answer if you draw a horizontal strip for the same problem.

Limits: x varies from y^2 to $\sqrt{y}$; y varies from 0 to 1. **Ans:** $I = \frac{1}{12}$.

■

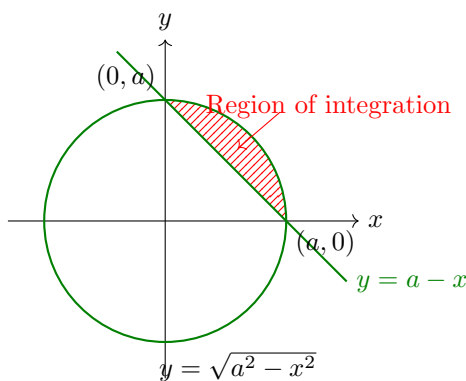
Problem 6.6

Evaluate $\int_0^a \int_{a-x}^{\sqrt{a^2-x^2}} y \, dx \, dy$ and also sketch the region of integration roughly.

Solution. Let $I = \int_0^a \int_{a-x}^{\sqrt{a^2-x^2}} y \, dx \, dy$, not in the proper order.

$$= \int_0^a \int_{a-x}^{\sqrt{a^2-x^2}} y \, dy \, dx \quad (1)$$

From the above, we see that y varies from $a - x$ to $\sqrt{a^2 - x^2}$ and x varies from $x = 0$ to $x = a$, i.e. y varies from the straight line $x + y = a$ (i.e. $\frac{x}{a} + \frac{y}{a} = 1$, a straight line with intercepts $(a, 0)$ and $(0, a)$) to the circle $x^2 + y^2 = a^2$ (a circle of centre at the origin $(0, 0)$ and radius a).



Now

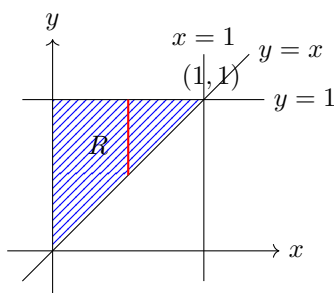
$$\begin{aligned}
 I &= \int_0^a \left(\int_{a-x}^{\sqrt{a^2-x^2}} y \, dy \right) dx = \int_0^a \left(\frac{y^2}{2} \right)_{a-x}^{\sqrt{a^2-x^2}} dx \\
 &= \frac{1}{2} \int_0^a ((a^2 - x^2) - (a - x)^2) dx \\
 &= \frac{1}{2} \left[a^2(x)_0^a - \left(\frac{x^3}{3} \right)_0^a - \left(\frac{(a-x)^3}{3(-1)} \right)_0^a \right] \\
 &= \frac{1}{2} \left[a^3 - \frac{a^3}{3} + \frac{1}{3}(0 - a^3) \right] \\
 &= \frac{1}{2} \left(a^3 - \frac{a^3}{3} - \frac{a^3}{3} \right) = \frac{1}{2} \left(a^3 - \frac{2a^3}{3} \right) = \frac{1}{2} \cdot \frac{a^3}{3} \Rightarrow \boxed{I = \frac{a^3}{6}}
 \end{aligned}$$

■

Problem 6.7

Evaluate $\iint_R \frac{y}{x^2 + y^2} dx dy$, where R is the region bounded by the straight lines $x = 0$ to $x = 1$ and $y = x$ to $y = 1$.

Solution. Let $I = \iint_R \frac{y}{x^2 + y^2} dx dy$ where R is the region bounded by $x = 0$ to $x = 1$ and $y = x$ to $y = 1$. $\therefore$ Based on this we have the following graph.



$\therefore$ In this region R , if we draw a vertical strip, y varies from $y = x$ to $y = 1$ and x varies from $x = 0$ to $x = 1$.

$$\therefore I = \iint_R \frac{y}{x^2 + y^2} dy dx = \int_{x=0}^1 \left(\int_{y=x}^1 \frac{y}{x^2 + y^2} dy \right) dx.$$

Let $t = x^2 + y^2$ (x is temporarily constant) $\Rightarrow dt = 2y dy$; when $y = x \Rightarrow t = 2x^2$; when $y = 1 \Rightarrow t = 1 + x^2$.

$$I = \int_0^1 \left(\int_{2x^2}^{1+x^2} \frac{dt/2}{t} \right) dx = \frac{1}{2} \int_0^1 \left(\int_{2x^2}^{1+x^2} \frac{dt}{t} \right) dx = \frac{1}{2} \int_0^1 (\log_e t)_{2x^2}^{1+x^2} dx$$

$$= \frac{1}{2} \int_0^1 (\log_e(1+x^2) - \log_e 2x^2) dx = \frac{1}{2} \int_0^1 \log_e \left(\frac{1+x^2}{2x^2} \right) dx = \frac{1}{2} \int u dv,$$

where $u = \log_e \left(\frac{x^2+1}{2x^2} \right)$ and $dv = dx$, so $v = x$ and

$$\begin{aligned} du &= \frac{1}{\frac{x^2+1}{2x^2}} \left[\frac{(2x^2)(2x) - (x^2+1)(4x)}{(2x^2)^2} \right] dx = \frac{2x^2}{x^2+1} \left(\frac{4x^3 - 4x^3 - 4x}{4x^4} \right) dx \\ &= -\frac{2x^2}{x^2+1} \left(\frac{4x}{4x^4} \right) dx \Rightarrow du = \frac{-2}{x(x^2+1)} dx. \end{aligned}$$

$$\begin{aligned} \therefore I &= \frac{1}{2} \left[uv - \int v du \right] = \frac{1}{2} \left\{ \left(\log_e \left(\frac{x^2+1}{2x^2} \right) x \right)_0^1 - \int_0^1 x \left(\frac{-2}{x(x^2+1)} \right) dx \right\} \\ &= \frac{1}{2} \left\{ [\log_e 1 - 0] + 2 \int_0^1 \frac{dx}{1+x^2} \right\} = \frac{1}{2} \left\{ 0 + 2 (\tan^{-1} x)_0^1 \right\} \\ &= \frac{1}{2} \left\{ 0 + 2 \left(\frac{\pi}{4} - 0 \right) \right\} = \frac{\pi}{4} \Rightarrow \boxed{I = \frac{\pi}{4}} \end{aligned}$$

(At the lower limit, $x \log_e \frac{x^2+1}{2x^2} \rightarrow 0$ as $x \rightarrow 0$.) ■

Exercises.

- (1) Evaluate $\iint xy \, dx \, dy$ over the region in the positive quadrant bounded by the line $2x+3y=6$ (i.e. $\frac{x}{3} + \frac{y}{2} = 1$, through $(3,0)$ and $(0,2)$).
- (2) Evaluate $\iint (x+y) \, dx \, dy$ over the region in the positive quadrant bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. (**Hint:** y varies from 0 to $\frac{b}{a}\sqrt{a^2-x^2}$, x from 0 to a .)

6.4 Change of order of integration

Under this, one can evaluate the given multiple integral by changing the order of integration, i.e. $dx \, dy$ to $dy \, dx$ and vice versa, i.e. suppose we have

$$I = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y) \, dy \, dx;$$

then by change of order of integration we can write

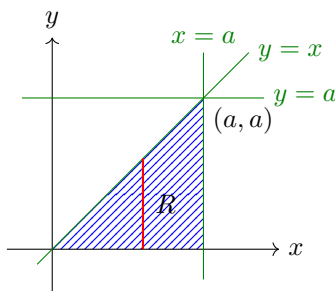
$$I = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) \, dx \, dy,$$

where the new limits are read off from a sketch of the same region, using a horizontal strip instead of a vertical strip (or vice versa).

Problem 6.8

Evaluate $\int_0^a \int_y^a \frac{x}{x^2+y^2} \, dx \, dy$ by change of order of integration.

Solution. Given $I = \int_{y=0}^a \int_{x=y}^a \frac{x}{x^2+y^2} \, dx \, dy$; it is in proper order. To change the order of integration, consider the limits: x varies from y to a ; y varies from 0 to a , i.e. $x = y$ to $x = a$ and $y = 0$ to $y = a$.



Under the given limits, the region of integration R is shaded. To change the order of integration, draw the vertical strip (already they have considered horizontal movement) in the same region. We get y varies from $y = 0$ to $y = x$ and x varies from $x = 0$ to $x = a$.

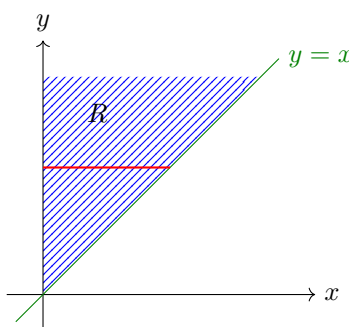
$$\begin{aligned}
 \therefore I &= \int_0^a \int_y^a \frac{x}{x^2 + y^2} dx dy \text{ becomes } \int_{x=0}^a \left(\int_{y=0}^x \frac{x}{x^2 + y^2} dy \right) dx \quad (\text{by change of order of integration}) \\
 &= \int_0^a x \left(\int_0^x \frac{dy}{x^2 + y^2} \right) dx = \int_0^a x \left(\frac{1}{x} \tan^{-1} \left(\frac{y}{x} \right) \right)_0^x dx \\
 &= \int_0^a \left(\tan^{-1} \left(\frac{x}{x} \right) - \tan^{-1} \left(\frac{0}{x} \right) \right) dx \\
 &= \int_0^a (\tan^{-1}(1) - \tan^{-1}(0)) dx = \int_0^a \frac{\pi}{4} dx = \frac{\pi}{4} (x)_0^a \Rightarrow \boxed{I = \frac{\pi a}{4}}
 \end{aligned}$$

■

Problem 6.9

Evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} dy dx$ by using change of order of integration.

Solution. Given $I = \int_{x=0}^\infty \int_{y=x}^\infty \frac{e^{-y}}{y} dy dx$, in proper order. To change the order of integration, consider the limits: x varies from $x = 0$ to $x = \infty$; y varies from $y = x$ to $y = \infty$.



Based on the given limits, the region of integration R is shaded, and they have considered the vertical movement in that region. To change the order of integration, draw a horizontal strip in that region; we get x varies from $x = 0$ to $x = y$ and y varies from $y = 0$ to $y = \infty$. $\therefore$ By change of order of integration, we get

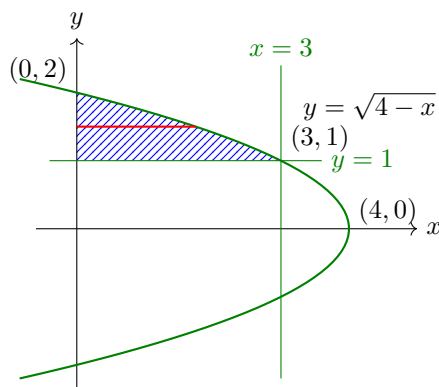
$$\begin{aligned}
 I &= \int_{y=0}^\infty \left(\int_{x=0}^y \frac{e^{-y}}{y} dx \right) dy = \int_0^\infty \frac{e^{-y}}{y} (x)_0^y dy = \int_0^\infty \frac{e^{-y}}{y} y dy = \int_0^\infty e^{-y} dy \\
 &= \left(\frac{e^{-y}}{-1} \right)_0^\infty = -(e^{-\infty} - e^0) = -(0 - 1) \Rightarrow \boxed{I = 1}
 \end{aligned}$$

■

Problem 6.10

Evaluate $\int_0^3 \int_1^{\sqrt{4-x}} (x+y) dy dx$ by change of order of integration.

Solution. Given $I = \int_0^3 \int_1^{\sqrt{4-x}} (x+y) dy dx$, proper order. To change the order: x varies from $x = 0$ to $x = 3$; y varies from $y = 1$ to $y = \sqrt{4-x}$, i.e. $y^2 = 4-x$ (a parabola through $(4, 0)$, $(0, \pm 2)$; at $x = 0$, $y = 2$; at $x = 3$, $y = 1$).



Under a horizontal strip, x varies from $x = 0$ to $x = 4 - y^2$ and y varies from $y = 1$ to 2 :

$$\begin{aligned}
 I &= \int_{y=1}^2 \int_{x=0}^{4-y^2} (x+y) dx dy = \int_1^2 \left[\frac{x^2}{2} + xy \right]_0^{4-y^2} dy \\
 &= \int_1^2 \left[\frac{(4-y^2)^2}{2} + y(4-y^2) \right] dy \\
 &= \int_1^2 \left[8 - 4y^2 + \frac{y^4}{2} + 4y - y^3 \right] dy \\
 &= \left[8y - \frac{4y^3}{3} + \frac{y^5}{10} + 2y^2 - \frac{y^4}{4} \right]_1^2 \\
 &= 8 - \frac{28}{3} + \frac{31}{10} + 6 - \frac{15}{4} \Rightarrow \boxed{I = \frac{241}{60}}
 \end{aligned}$$

■

7 Triple integration

Integrals of a function $f(x, y, z)$ in $\mathbb{R}^3$ are called triple integrals. Evaluation of a triple integral takes the following order:

$$\int_{x=a}^b \left(\int_{y=\phi_1(x)}^{\phi_2(x)} \left(\int_{z=\psi_1(x,y)}^{\psi_2(x,y)} f(x, y, z) dz \right) dy \right) dx.$$

Problem 7.1

Evaluate $\int_0^2 \int_1^3 \int_1^2 xy^2 z dz dy dx$.

Solution.

$$I = \int_0^2 \int_1^3 \int_1^2 xy^2 z dz dy dx = \int_0^2 \int_1^3 xy^2 \left(\frac{z^2}{2} \right)_1^2 dy dx = \int_0^2 \int_1^3 xy^2 \left(2 - \frac{1}{2} \right) dy dx$$

$$\begin{aligned}
&= \frac{3}{2} \int_0^2 x \left(\frac{y^3}{3} \right)_1^3 dx = \frac{3}{2} \int_0^2 x \left(9 - \frac{1}{3} \right) dx = \frac{3}{2} \times \frac{26}{3} \int_0^2 x dx \\
&= \frac{3}{2} \times \frac{26}{3} \left(\frac{x^2}{2} \right)_0^2 = \frac{3}{2} \times \frac{26}{3} \times 2 \\
&\Rightarrow \boxed{I = 26}
\end{aligned}$$

■

Problem 7.2

Evaluate $\int_0^{\log_e 2} \int_0^{2x} \int_0^{x+y} e^{x+y+z} dx dy dz$.

Solution. Given (not in proper order; the correct order is $dz dy dx$)

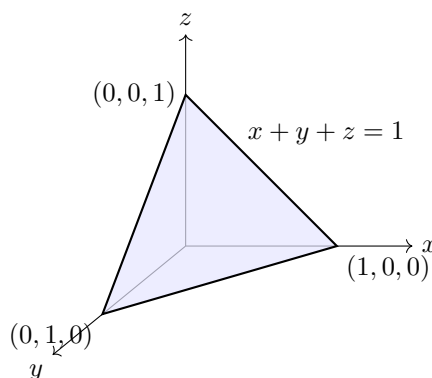
$$\begin{aligned}
I &= \int_0^{\log 2} \left(\int_0^{2x} \left(\int_0^{x+y} e^{(x+y+z)} dz \right) dy \right) dx \\
&= \int_0^{\log 2} \int_0^{2x} \left(e^{(x+y)+z} \right)_0^{x+y} dy dx \\
&= \int_0^{\log 2} \int_0^{2x} \left(e^{(x+y)+(x+y)} - e^{(x+y)+0} \right) dy dx \\
&= \int_0^{\log 2} \left(\int_0^{2x} (e^{2x+2y} - e^{x+y}) dy \right) dx \\
&= \int_0^{\log 2} \left\{ \left(\frac{e^{2x+2y}}{2} \right)_0^{2x} - (e^{x+y})_0^{2x} \right\} dx \\
&= \int_0^{\log 2} \left\{ \frac{1}{2} (e^{6x} - e^{2x}) - (e^{3x} - e^x) \right\} dx \\
&= \left(\frac{e^{6x}}{12} \right)_0^{\log 2} - \left(\frac{e^{2x}}{4} \right)_0^{\log 2} - \left(\frac{e^{3x}}{3} \right)_0^{\log 2} + (e^x)_0^{\log 2} \\
&= \frac{1}{12} (e^{6 \log 2} - 1) - \frac{1}{4} (e^{2 \log 2} - 1) - \frac{1}{3} (e^{3 \log 2} - 1) + (e^{\log 2} - 1) \\
&= \frac{1}{12} (2^6 - 1) - \frac{1}{4} (2^2 - 1) - \frac{1}{3} (2^3 - 1) + (2 - 1) = \frac{1}{12} (63) - \frac{1}{4} (3) - \frac{1}{3} (7) + 1 \\
&= \frac{1}{12} (63 - 9 - 28 + 12) = \frac{1}{12} (38) = \frac{19}{6} \Rightarrow \boxed{I = \frac{19}{6}}
\end{aligned}$$

■

Problem 7.3

Evaluate $\iiint_R \frac{dx dy dz}{(x+y+z+1)^3}$, where R is the region bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$.

Solution. Given $I = \iiint_R \frac{dx dy dz}{(x+y+z+1)^3}$ where R is the region as shown in the figure, based on the given limits.



In this region R , we have z varies from $\boxed{z = 0 \text{ to } 1 - x - y}$. To find the limits of x and y , project the surface $x + y + z = 1$ onto the xy -plane by putting $z = 0 \Rightarrow$ the surface equation is $x + y = 1$. In this surface, if we draw a vertical strip, y varies from $\boxed{y = 0 \text{ to } 1 - x}$ and x varies from $\boxed{x = 0 \text{ to } 1}$.

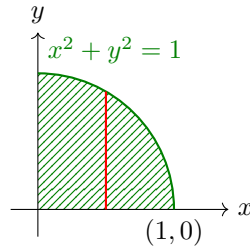
$$\begin{aligned}
 \therefore I &= \int_0^1 \left(\int_0^{1-x} \left(\int_0^{1-x-y} (x+y+z+1)^{-3} dz \right) dy \right) dx \\
 &= \int_0^1 \left(\int_0^{1-x} \left[\frac{(x+y+1+z)^{-2}}{-2} \right]_0^{1-x-y} dy \right) dx \\
 &= -\frac{1}{2} \int_0^1 \int_0^{1-x} \left\{ ((x+y+1) + (1-x-y))^{-2} - (x+y+1)^{-2} \right\} dy dx \\
 &= -\frac{1}{2} \int_0^1 \left(\int_0^{1-x} \left(\frac{1}{4} - (x+y+1)^{-2} \right) dy \right) dx \\
 &= -\frac{1}{2} \int_0^1 \left\{ \frac{1}{4} (y)_0^{1-x} - \left(\frac{((x+1)+y)^{-1}}{-1} \right)_0^{1-x} \right\} dx \\
 &= -\frac{1}{2} \int_0^1 \left\{ \frac{1}{4} (1-x) + \left[((x+1) + (1-x))^{-1} - (x+1+0)^{-1} \right] \right\} dx \\
 &= -\frac{1}{2} \int_0^1 \left\{ \frac{1}{4} (1-x) + \left(\frac{1}{2} - \frac{1}{1+x} \right) \right\} dx \\
 &= -\frac{1}{2} \left(\frac{1}{4} \left(\frac{(1-x)^2}{2(-1)} \right)_0^1 + \frac{1}{2} (x)_0^1 - (\log_e(1+x))_0^1 \right) \\
 &= -\frac{1}{2} \left[-\frac{1}{8} (0-1) + \frac{1}{2} (1) - (\log_e 2 - \log_e 1) \right] = -\frac{1}{2} \left(\frac{1}{8} + \frac{1}{2} - \log_e 2 \right) \quad (\because \log 1 = 0) \\
 \Rightarrow \boxed{I = \frac{1}{2} \left(\log_e 2 - \frac{5}{8} \right)}
 \end{aligned}$$

■

Problem 7.4

Evaluate $\iiint_R \frac{dz dy dx}{\sqrt{1-x^2-y^2-z^2}}$, where R is the region bounded by the positive octant of the sphere $x^2 + y^2 + z^2 = 1$.

Solution. Given $I = \iiint_R \frac{dz dy dx}{\sqrt{1-x^2-y^2-z^2}}$, where R is the region shown in the figure, in which z varies from $\boxed{z = 0 \text{ to } \sqrt{1-x^2-y^2}}$.



To get the x and y limits, project the surface onto the xy -plane by putting $z = 0 \Rightarrow$ we get the equation $x^2 + y^2 = 1$, a circle with centre $(0, 0)$ and radius 1, in the positive quadrant. In this region, if we draw a vertical strip, we get y varies from $y = 0$ to $\sqrt{1 - x^2}$ and x varies from $x = 0$ to 1 .

$$\begin{aligned}
 \therefore I &= \int_0^1 \left(\int_0^{\sqrt{1-x^2}} \left(\int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{(1-x^2-y^2)-z^2}} dz \right) dy \right) dx \\
 &= \int_0^1 \int_0^{\sqrt{1-x^2}} \left(\int_0^{\sqrt{1-x^2-y^2}} \frac{dz}{\sqrt{(\sqrt{1-x^2-y^2})^2 - z^2}} \right) dy dx \quad \left(\because \int \frac{dz}{\sqrt{a^2 - z^2}} = \sin^{-1} \frac{z}{a} \right) \\
 &= \int_0^1 \int_0^{\sqrt{1-x^2}} \left[\sin^{-1} \left(\frac{z}{\sqrt{1-x^2-y^2}} \right) \right]_0^{\sqrt{1-x^2-y^2}} dy dx \\
 &= \int_0^1 \left(\int_0^{\sqrt{1-x^2}} (\sin^{-1}(1) - \sin^{-1}(0)) dy \right) dx \\
 &= \int_0^1 \int_0^{\sqrt{1-x^2}} \left(\frac{\pi}{2} \right) dy dx = \int_0^1 \frac{\pi}{2} (y)_0^{\sqrt{1-x^2}} dx = \int_0^1 \frac{\pi}{2} (\sqrt{1-x^2}) dx \\
 &= \frac{\pi}{2} \int_0^1 \sqrt{1-x^2} dx \\
 &\quad \left(\because \int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right) \\
 &= \frac{\pi}{2} \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1}(x) \right]_0^1 = \frac{\pi}{2} \left[(0-0) + \frac{1}{2} \left(\frac{\pi}{2} - 0 \right) \right] \Rightarrow \boxed{I = \frac{\pi^2}{8}}
 \end{aligned}$$

■

8 Applications: area, volume, mass and moments of inertia

8.1 Area of a region R using double integration

Result 8.1 (Area)

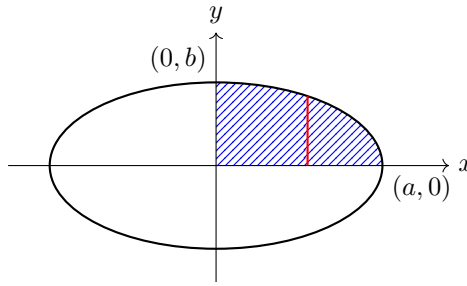
The area of any region bounded by the curves is given by

$$\text{Area} = \iint_R dx dy = \iint_R dy dx.$$

Problem 8.1

Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ using double integration.

Solution. Given $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{y^2}{b^2} = 1 - \frac{x^2}{a^2} \Rightarrow y^2 = \frac{b^2}{a^2}(a^2 - x^2) \Rightarrow y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$.



If R is the positive quadrant region of the given ellipse, then $\text{Area} = 4 \iint_R dx dy$ (1). In this region R , if we draw a vertical strip, then we have y varies from $y = 0$ to $\frac{b}{a} \sqrt{a^2 - x^2}$ and x varies from $x = 0$ to a .

$$\begin{aligned}
 \therefore (1) \Rightarrow \text{Area} &= 4 \int_0^a \left(\int_0^{\frac{b}{a} \sqrt{a^2 - x^2}} dy \right) dx = 4 \int_0^a (y)_0^{\frac{b}{a} \sqrt{a^2 - x^2}} dx \\
 &= 4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx = \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} dx \\
 &= \frac{4b}{a} \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right]_0^a \\
 &= \frac{4b}{a} \left[(0 - 0) + \frac{a^2}{2} (\sin^{-1}(1) - \sin^{-1}(0)) \right] \\
 &= \frac{4b}{a} \left(\frac{a^2}{2} \cdot \frac{\pi}{2} \right) \Rightarrow \boxed{\text{Area of the ellipse} = \pi ab}
 \end{aligned} \tag{2}$$

Aliter. We can solve equation (2), i.e. $\int_0^a \sqrt{a^2 - x^2} dx$, using parametric form substitution also. Let $x = a \cos \theta$, $dx = -a \sin \theta d\theta$; when $x = 0$, $\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$; when $x = a$, $\cos \theta = 1 \Rightarrow \theta = 0$.

$$\begin{aligned}
 I &= \int_0^a \sqrt{a^2 - x^2} dx = \int_{\pi/2}^0 \sqrt{a^2 - a^2 \cos^2 \theta} (-a \sin \theta) d\theta = \int_0^{\pi/2} \sqrt{a^2 \sin^2 \theta} a \sin \theta d\theta \\
 &= a^2 \int_0^{\pi/2} \sin^2 \theta d\theta = a^2 \frac{\sqrt{\pi}}{2} \frac{\Gamma(\frac{3}{2})}{\Gamma(\frac{4}{2})} \quad \left(\because \int_0^{\pi/2} \sin^p \theta d\theta = \frac{\sqrt{\pi}}{2} \frac{\Gamma(\frac{p+1}{2})}{\Gamma(\frac{p+2}{2})} \right) \\
 &= \frac{a^2 \sqrt{\pi}}{2} \frac{\frac{1}{2} \Gamma(\frac{1}{2})}{1} \quad (\because \Gamma(2) = 1!, \Gamma(\frac{3}{2}) = \Gamma(\frac{1}{2} + 1)) \\
 &= \frac{a^2 \sqrt{\pi}}{2} \cdot \frac{1}{2} \sqrt{\pi} = \frac{\pi a^2}{4}.
 \end{aligned}$$

$\therefore$ By (2), $\text{Area} = \frac{4b}{a} \int_0^a \sqrt{a^2 - x^2} dx = \frac{4b}{a} \cdot \frac{\pi a^2}{4} = \boxed{\pi ab}$. ■

Problem 8.2

Find the area of the circle $x^2 + y^2 = a^2$ using double integration.

Solution. Given $x^2 + y^2 = a^2 \Rightarrow y^2 = a^2 - x^2 \Rightarrow y = \pm \sqrt{a^2 - x^2}$. Under a vertical strip in the positive quadrant, y varies from 0 to $\sqrt{a^2 - x^2}$ and x from 0 to a :

$$\text{Area} = 4 \iint_R dy dx = 4 \int_0^a \int_0^{\sqrt{a^2 - x^2}} dy dx = 4 \int_0^a \sqrt{a^2 - x^2} dx = 4 \cdot \frac{\pi a^2}{4} = \boxed{\pi a^2}$$

(using the previous problem; equivalently, put $b = a$ in πab). ■

Problem 8.3

Find the area of the region R bounded by the curves $y = x^2$ and $y^2 = x$.

Solution. Given R is bounded by $y = x^2$ and $y^2 = x$ (see the figure in the earlier problem; they intersect at $(0, 0)$ and $(1, 1)$). Under a horizontal strip, x varies from $x = y^2$ to $x^2 = y$, i.e. $x = y^2$ to $\sqrt{y}$, and y varies from $y = 0$ to 1 .

$$\begin{aligned}\therefore \text{Area of } R &= \iint_R dx dy = \int_0^1 \left(\int_{y^2}^{\sqrt{y}} dx \right) dy = \int_0^1 (x)_{y^2}^{\sqrt{y}} dy = \int_0^1 (\sqrt{y} - y^2) dy \\ &= \left(\frac{y^{3/2}}{3/2} \right)_0^1 - \left(\frac{y^3}{3} \right)_0^1 = \frac{2}{3} - \frac{1}{3} \Rightarrow \boxed{\text{Area of } R = \frac{1}{3}}\end{aligned}$$

■

8.2 Volume of a solid using triple integration

Result 8.2 (Volume)

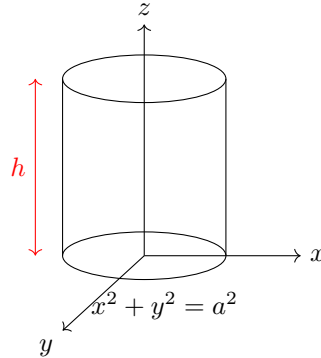
The volume of any solid bounded by the planes (surfaces) in $\mathbb{R}^3$ is given by

$$\text{Volume} = \iiint_V dx dy dz.$$

Problem 8.4

Find the volume of the cylinder $x^2 + y^2 = a^2$, $z = 0$ and $z = h$ using triple integration.

Solution. Given the cylinder $x^2 + y^2 = a^2$ and $z = 0$, $z = h$.



Based on the given equations, we have $z = 0$ to h , $y = -\sqrt{a^2 - x^2}$ to $\sqrt{a^2 - x^2}$, $x = -a$ to a .

$$\begin{aligned}\therefore \text{Volume of the cylinder} &= \iiint_V dz dy dx = \int_{-a}^a \left(\int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \left(\int_0^h dz \right) dy \right) dx \\ &= \int_{-a}^a \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} (z)_0^h dy dx \\ &= h \int_{-a}^a (y)_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dx = h \int_{-a}^a \left(\sqrt{a^2-x^2} - \left(-\sqrt{a^2-x^2} \right) \right) dx \\ &= 2h \int_{-a}^a \sqrt{a^2-x^2} dx = 2h \cdot 2 \int_0^a \sqrt{a^2-x^2} dx \quad (\because \text{even function}) \\ &= 4h \cdot \frac{\pi a^2}{4} \quad (\because \text{by Problem 1}) = \boxed{\pi a^2 h} = \text{volume of the cylinder.}\end{aligned}$$

■

Problem 8.5

Find the volume of the tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$.

Solution. Based on the given equations (see the figure of the tetrahedron above), z varies from $z = 0$ to $1 - x - y$, y varies from $y = 0$ to $1 - x$, x varies from $x = 0$ to 1 .

$$\begin{aligned} \therefore \text{Volume of the tetrahedron} &= \int_0^1 \left(\int_0^{1-x} \left(\int_0^{1-x-y} dz \right) dy \right) dx = \int_0^1 \int_0^{1-x} (1 - x - y) dy dx \\ &= \int_0^1 \left[(1-x)y - \frac{y^2}{2} \right]_0^{1-x} dx = \int_0^1 \frac{(1-x)^2}{2} dx = \frac{1}{2} \left[\frac{(1-x)^3}{-3} \right]_0^1 = \frac{1}{6} \end{aligned}$$

■

Problem 8.6

Find the volume of the cuboid bounded by the planes $x = 0$, $x = a$; $y = 0$, $y = b$; $z = 0$ and $z = c$.

Solution. Here x varies from $x = 0$ to a , y varies from $y = 0$ to b , z varies from $z = 0$ to c .

$$\text{Volume of the cuboid} = \int_0^a \left(\int_0^b \left(\int_0^c dz \right) dy \right) dx = \int_0^a \int_0^b c dy dx = \int_0^a bc dx \Rightarrow \text{Volume} = abc$$

■

8.3 Mass and moments of inertia

Using double and triple integration we have the following.

Result 8.3 (Under double integration)

In a region R with density function $f(x, y)$:

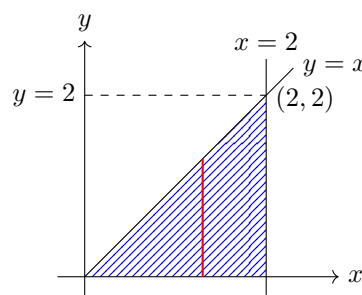
- (1) **Area** of the region R : $\text{Area} = \iint_R dx dy$.
- (2) **Mass** m of R : $m = \iint_R f(x, y) dx dy$.
- (3) **Moments of inertia** about the x -axis (I_x), the y -axis (I_y) and the origin (I_0):

$$I_x = \iint_R y^2 f(x, y) dx dy, \quad I_y = \iint_R x^2 f(x, y) dx dy, \quad I_0 = I_x + I_y = \iint_R (x^2 + y^2) f(x, y) dx dy.$$

Problem 8.7 (Example)

Find the mass m of the surface bounded by $x = 0$, $x = 2$, $y = 0$ and $y = x$ with density $f(x, y) = xy$. Also find the moment of inertia about the x -axis, the y -axis and the origin.

Solution. Given $f(x, y) = xy$, and the region R is bounded by $x = 0$, $x = 2$, $y = 0$ and $y = x$. $\therefore$ We have the following figure.



If we draw a vertical strip in this region, we have y varies from $y = 0$ to $y = x$ and x varies from $x = 0$ to $x = 2$. (On the other hand, if we draw a horizontal strip in this region, we get x varies from $x = y$ to $x = 2$ and y varies from $y = 0$ to $y = 2$.)

Hence the mass

$$\begin{aligned} m &= \iint_R f(x, y) dx dy = \int_{x=0}^2 \int_{y=0}^x xy dy dx \quad (\because \text{vertical strip condition; } y \text{ takes variable limits}) \\ &= \int_0^2 x \left(\frac{y^2}{2} \right)_0^x dx = \frac{1}{2} \int_0^2 x(x^2 - 0) dx = \frac{1}{2} \int_0^2 x^3 dx = \frac{1}{2} \left(\frac{x^4}{4} \right)_0^2 \Rightarrow \boxed{m = 2 \text{ units}} \end{aligned}$$

Moment of inertia about the x -axis:

$$\begin{aligned} I_x &= \iint_R y^2 f(x, y) dx dy = \int_0^2 \int_0^x y^2 xy dy dx = \int_0^2 \int_0^x xy^3 dy dx = \int_0^2 x \left(\frac{y^4}{4} \right)_0^x dx \\ &= \frac{1}{4} \int_0^2 x^5 dx = \frac{1}{4} \left(\frac{x^6}{6} \right)_0^2 = \frac{1}{4} \left(\frac{64}{6} \right) \Rightarrow \boxed{I_x = \frac{16}{6} = \frac{8}{3}} \end{aligned}$$

Similarly, about the y -axis:

$$I_y = \iint_R x^2 f(x, y) dx dy = \int_0^2 \int_0^x x^2(xy) dy dx = \int_0^2 \int_0^x x^3 y dy dx = \int_0^2 \frac{x^5}{2} dx = \frac{64}{12} \Rightarrow \boxed{I_y = \frac{16}{3}}$$

$\therefore$ About the origin:

$$\boxed{I_0 = I_x + I_y = \frac{8}{3} + \frac{16}{3} = \frac{24}{3} = 8}$$

$$(OR) I_0 = \iint_R (x^2 + y^2) f(x, y) dx dy = \int_0^2 \int_0^x (x^2 + y^2)(xy) dy dx = 8. \quad \blacksquare$$

Result 8.4 (Under triple integration)

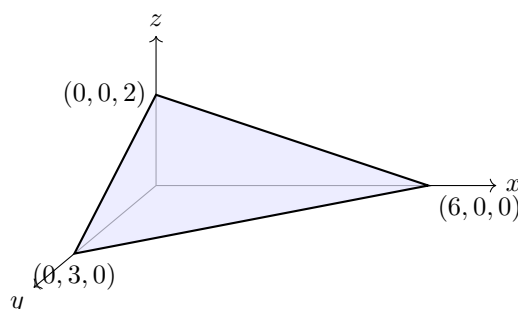
For a solid S bounded by volume V , with density $f(x, y, z)$ on S :

- (1) **Volume** $V = \iiint_S dx dy dz$.
- (2) **Mass** m of S : $m = \iiint_S f(x, y, z) dx dy dz$.
- (3) **Moments of inertia:**
 - (i) about the yz -plane: $I_x = \iiint_S (y^2 + z^2) f(x, y, z) dx dy dz$;
 - (ii) about the zx -plane: $I_y = \iiint_S (z^2 + x^2) f(x, y, z) dx dy dz$;
 - (iii) about the xy -plane: $I_z = \iiint_S (x^2 + y^2) f(x, y, z) dx dy dz$.

Problem 8.8 (Example)

Suppose a solid S is bounded by the coordinate planes and the plane $x + 2y + 3z = 6$, with density $f(x, y, z) = x^2yz$. Find (i) the mass m of S ; (ii) the moments of inertia about the yz , zx and xy planes; (iii) the volume of the tetrahedron.

Solution. Given the surface $x + 2y + 3z = 6 \Rightarrow \frac{x}{6} + \frac{y}{3} + \frac{z}{2} = 1$, a tetrahedron with vertices $(6, 0, 0)$, $(0, 3, 0)$, $(0, 0, 2)$ as shown in the figure.



Here z varies from $z = 0$ to $z = \frac{1}{3}(6 - x - 2y)$. Put $z = 0$ in the surface equation; we can project onto the xy -plane. In the xy -plane the surface equation is $x + 2y = 6$ ($\because$ put $z = 0$). Then y varies from $y = 0$ to $\frac{1}{2}(6 - x)$ and x varies from $x = 0$ to $x = 6$. So the limits are

$$z = 0 \text{ to } \frac{1}{3}(6 - x - 2y), \quad y = 0 \text{ to } \frac{1}{2}(6 - x), \quad x = 0 \text{ to } 6.$$

(i) **Mass.**

$$m = \iiint_S f(x, y, z) \, dx \, dy \, dz = \int_{x=0}^6 \int_{y=0}^{\frac{1}{2}(6-x)} \int_{z=0}^{\frac{1}{3}(6-x-2y)} x^2 y z \, dz \, dy \, dx = \boxed{\frac{108}{35}}$$

(Integrating in z first: $\int_0^{\frac{1}{3}(6-x-2y)} z \, dz = \frac{(6-x-2y)^2}{18}$, so $m = \frac{1}{18} \int_0^6 x^2 \int_0^{\frac{1}{2}(6-x)} y(6-x-2y)^2 \, dy \, dx$; with $u = 6 - x$, the inner integral is $\int_0^{u/2} y(u-2y)^2 \, dy = \frac{u^4}{48}$, hence $m = \frac{1}{864} \int_0^6 x^2(6-x)^4 \, dx = \frac{1}{864} \cdot 6^7 \beta(3, 5) = \frac{108}{35}$.)

(ii) **Moments of inertia.**

(a) About the yz -plane:

$$I_x = \iiint_S (y^2 + z^2) f(x, y, z) \, dx \, dy \, dz = \int_0^6 \int_0^{\frac{1}{2}(6-x)} \int_0^{\frac{1}{3}(6-x-2y)} (y^2 + z^2)(x^2 y z) \, dz \, dy \, dx = \boxed{\frac{117}{35}}$$

(b) About the zx -plane:

$$I_y = \iiint_S (z^2 + x^2) f(x, y, z) \, dx \, dy \, dz = \int_0^6 \int_0^{\frac{1}{2}(6-x)} \int_0^{\frac{1}{3}(6-x-2y)} (z^2 + x^2)(x^2 y z) \, dz \, dy \, dx = \boxed{\frac{684}{35}}$$

(c) About the xy -plane:

$$I_z = \iiint_S (x^2 + y^2) f(x, y, z) \, dx \, dy \, dz = \int_0^6 \int_0^{\frac{1}{2}(6-x)} \int_0^{\frac{1}{3}(6-x-2y)} (x^2 + y^2)x^2 y z \, dz \, dy \, dx = \boxed{\frac{729}{35}}$$

(iii) **Volume of the tetrahedron.**

$$\begin{aligned} V &= \iiint_S dx \, dy \, dz = \int_{x=0}^6 \int_{y=0}^{\frac{1}{2}(6-x)} \int_{z=0}^{\frac{1}{3}(6-x-2y)} dz \, dy \, dx = \int_0^6 \int_0^{\frac{1}{2}(6-x)} (z)_0^{\frac{1}{3}(6-x-2y)} dy \, dx \\ &= \int_0^6 \left(\int_0^{\frac{1}{2}(6-x)} \frac{1}{3}(6-x-2y) \, dy \right) dx \\ &= \frac{1}{3} \int_0^6 \left[6(y)_0^{\frac{1}{2}(6-x)} - x(y)_0^{\frac{1}{2}(6-x)} - 2 \left(\frac{y^2}{2} \right)_0^{\frac{1}{2}(6-x)} \right] dx \\ &= \frac{1}{3} \int_0^6 \left[6 \cdot \frac{1}{2}(6-x) - x \cdot \frac{1}{2}(6-x) - \frac{1}{4}(6-x)^2 \right] dx \\ &= \frac{1}{3} \int_0^6 \left((18-3x) - \left(3x - \frac{x^2}{2} \right) - \frac{1}{4}(36+x^2-12x) \right) dx \\ &= \frac{1}{3} \int_0^6 \left[x^2 \left(\frac{1}{2} - \frac{1}{4} \right) + x(-3-3+3) + (18-9) \right] dx = \frac{1}{3} \int_0^6 \left(\frac{1}{4}x^2 - 3x + 9 \right) dx \\ &= \frac{1}{3} \left[\frac{1}{12}(x^3)_0^6 - \frac{3}{2}(x^2)_0^6 + 9(x)_0^6 \right] = \frac{1}{3} \left[\frac{216}{12} - 54 + 54 \right] = \frac{1}{3}(18) \Rightarrow \boxed{\text{Volume} = 6 \text{ units}} \end{aligned}$$

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Assignment – IV

Under Beta–Gamma integrals

(1) Prove that $\int_0^1 \frac{x^{2a}}{\sqrt{1-x^2}} \, dx = \frac{\sqrt{\pi}}{2} \frac{\Gamma(a + \frac{1}{2})}{\Gamma(a + 1)}.$

- (2) Prove that $\int_0^\infty \sqrt{x} e^{-4\sqrt{x}} dx = \frac{1}{16}$.
- (3) Prove that $\int_0^1 \frac{dx}{\sqrt{\log \frac{1}{x}}} = \sqrt{\pi}$.
- (4) Prove that $\int_0^\infty \frac{e^{-x^2}}{\sqrt{x}} dx \int_0^\infty x^2 e^{-x^4} dx = \frac{\pi}{4\sqrt{2}}$.
- (5) Prove that $\int_0^\infty \frac{x^n}{n^x} dx = \frac{\Gamma(n+1)}{(\log_e n)^{n+1}}$.
- (6) Prove that $\int_0^1 \frac{x^2 dx}{\sqrt{1-x^4}} \cdot \int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\pi}{4}$.
- (7) Prove the following by taking $x^2 = \tan \theta$:
- (i) $\int_0^\infty \frac{dx}{1+x^4} = \frac{\pi}{2\sqrt{2}}$ (ii) $\int_0^\infty \frac{x^2 dx}{(1+x^4)^2} = \frac{\pi}{8\sqrt{2}}$ (iii) $\int_0^\infty \frac{x^2 dx}{(1+x^4)^3} = \frac{5\pi\sqrt{2}}{128}$.
- (8) Prove that $\int_0^{\pi/2} \frac{\cos^{2m-1} \theta \sin^{2n-1} \theta}{(a \cos^2 \theta + b \sin^2 \theta)^{m+n}} d\theta = \frac{\beta(m, n)}{2a^m b^n}$.
- (9) Prove that $\frac{\beta(m, n+1)}{n} = \frac{\beta(m+1, n)}{m} = \frac{\beta(m, n)}{m+n}$.
- (10) Prove that $\int_0^\infty \frac{x^{n-1}}{1+x} dx = \Gamma(n)\Gamma(1-n)$.

Under multiple integrals

- (1) Evaluate the following integrals using change of order of integration:
- (i) $\int_0^a \int_{\frac{b}{a}\sqrt{a^2-x^2}}^{\sqrt{a^2-y^2}} x^2 dy dx$ (ii) $\int_0^a \int_{a-y}^{\sqrt{a^2-y^2}} y dx dy$ (iii) $\int_0^a \int_{a-\sqrt{a^2-y^2}}^{a+\sqrt{a^2-y^2}} xy dx dy$
- (iv) $\int_0^4 \int_{x^2/4}^{\sqrt{x}} dy dx$ (v) $\int_1^2 \int_0^{1/x} xy dx dy$ (region bounded by the hyperbola $xy = 1$).
- (2) Find the area of the region bounded by $x = 0$, $x = 4$, $x^2 = 4y$ and $y^2 = x$ using double integration.
- (3) Evaluate the following:
- (i) $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx$ (ii) $\int_0^{2\pi} \int_0^\pi \int_0^a r^4 \sin \phi dr d\phi d\theta$ (iii) $\int_0^a \int_0^x \int_0^y xyz dz dy dx$.
- (4) Evaluate $\iiint_S xyz dx dy dz$, where S is the surface bounded by the positive octant of the sphere $x^2 + y^2 + z^2 = a^2$.
- (5) Find the volume of the solid bounded by the planes $x = 0$, $y = 0$, $z = 0$, $2x + 3y + 3z = 6$.
- (6) Find the mass at a point (x, y) of the region R , and also the moments of inertia about the x -axis, the y -axis and the origin, for the density function $f(x, y) = xy$ over the region R bounded by the lines $x = 0$, $y = 0$, $x + 2y = 6$.
- (7) Find the mass m over the solid S with density function $f(x, y, z) = xz$, where S is bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 2$. Also find the moments of inertia about the xy -plane, the yz -plane, the zx -plane and about the origin.